Tutorial 5 — Potential Outcomes, Causality, and Counterfactual Reasoning

Questions

Introductory Econometrics — TA Session

Duration: 50 minutes

Based on Wooldridge, Sections 1-4 and 2-7a

Block 1: Motivation — Why Do We Need Potential Outcomes? (5 min)

Causality, Ceteris Paribus, and Counterfactual Reasoning (Section 1-4)

In economics we want to know whether one variable has a **causal effect** on another. This requires **ceteris paribus** reasoning: what happens to an outcome when we change one factor, *holding all other factors fixed.* Since we can rarely hold other factors fixed, we resort to **counterfactual reasoning**: "what *would have happened* under a different state of the world?" The challenge is that we can never observe the same unit in both states simultaneously. This is the **fundamental problem of causal inference**. The textbook illustrates this with four examples:

- **Ex. 1.3 — Fertilizer on Crop Yield.** A farmer obtains 180 bushels/acre using fertilizer. The counterfactual (yield *without* fertilizer on the same plot, same season) is unobservable. An ideal experiment randomly assigns fertilizer to identical plots. In practice, farmers choose fertilizer based on soil quality — a **confounding factor** that also affects yield.

- **Ex. 1.4 — Return to Education.** If a person gets one more year of education, by how much does their wage rise? A social planner could randomly assign education levels, but this is infeasible. People *choose* education based on ability and background, so comparing wages across education levels confounds the education effect with pre-existing differences (**selection bias**).

- **Ex. 1.5 — Police and Crime.** Does hiring more police reduce crime? Cities with high crime already hire more police, creating a positive correlation even if police *reduce* crime. This is **reverse causality** (simultaneity).

- **Ex. 1.6 — Minimum Wage and Unemployment.** Political and economic forces that set the minimum wage also affect employment. It is impossible to isolate the causal effect without holding these **confounding forces** fixed.

In each case, the core problem is the same: we observe one state of the world but need to compare it with a counterfactual we cannot see. The **potential outcomes framework** (Section 2-7a) formalizes this problem mathematically, and **random assignment** provides the cleanest solution.

Question 1 (Quick warm-up)

(a) In Example 1.4, why can we *not* simply compare the average wage of people with 16 years of education to that of people with 15 years and call the difference the causal return to education?

(b) For Examples 1.3–1.6, each describes an ideal experiment that is infeasible. What is the *one* feature all these ideal experiments share that would solve the causal inference problem?

Block 2: Potential Outcomes Notation (13 min)

Section 2-7a — Counterfactual (or Potential) Outcomes, Causality, and Policy Analysis

Consider a binary treatment: $x_i = 1$ if unit i is in the treatment group, $x_i = 0$ if in the control group. For each unit i there are two **potential outcomes**: $y_i(1)$ (outcome if treated) and $y_i(0)$ (outcome if not treated). Both exist conceptually, but we only observe one.

The **individual treatment effect** is $\tau_i = y_i(1) - y_i(0)$.

The **Average Treatment Effect (ATE)** and **Average Treatment Effect on the Treated (ATT)** are:

$$ATE = E[y(1)] - E[y(0)] \tag{2.75}$$

$$ATT = E[y(1) - y(0) \mid x = 1] \tag{2.76}$$

The **observed outcome** is given by the "switching equation":

$$y_i = x_i \cdot y_i(1) + (1 - x_i) \cdot y_i(0) \tag{2.77}$$

Given a random sample, we observe only one of $y_i(0)$ and $y_i(1)$: the treatment "switches" which potential outcome we see.

Question 2 (Notation — Definitions)

Consider a binary treatment $x_i \in \{0, 1\}$ for individual i.

(a) Define the two **potential outcomes** $y_i(0)$ and $y_i(1)$. What do they represent?

(b) Define the **individual treatment effect** τ_i. Why can we never compute it directly?

(c) Write the **observed outcome** y_i in terms of x_i, $y_i(0)$, and $y_i(1)$ (equation [2.77]). Verify the formula for each value of x_i.

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Question 3 (Math — Algebraic manipulation)

Starting from $y_i = x_i \cdot y_i(1) + (1 - x_i) \cdot y_i(0)$:

(a) Show that this is equivalent to:

$$y_i = y_i(0) + \big[y_i(1) - y_i(0)\big] \cdot x_i = y_i(0) + \tau_i \cdot x_i$$

(b) Now suppose the treatment effect is **constant**: $\tau_i = \tau$ for all i. Let $\beta_0 = E[y(0)]$ and $u_i = y_i(0) - E[y(0)]$. Show that:

$$y_i = \beta_0 + \tau \cdot x_i + u_i$$

What does this equation look like? What is the connection to regression?

(c) Write the formulas for the **ATE** (eq. [2.75]) and the **ATT** (eq. [2.76]). In one sentence each, explain what population each one refers to.

Block 3: ATE vs. ATT and Selection Bias — Numerical Exercise (13 min)

Question 4 (Numerical exercise)

Suppose we have a population with two types of individuals:

Type	Fraction of population	$y_i(0)$	$y_i(1)$
A	0.6	2	5
B	0.4	4	6

(a) Compute the individual treatment effect τ for each type.

(b) Compute the **ATE**.

(c) Now suppose that **only Type A** individuals get treated ($x = 1$ for Type A, $x = 0$ for Type B). Compute the **ATT**. Is it equal to the ATE? Why or why not?

(d) Compute the **selection bias** $E[y(0) \mid x{=}1] - E[y(0) \mid x{=}0]$. What is its sign? Interpret.

(e) Verify the decomposition: compute both sides of

$$E[y \mid x{=}1] - E[y \mid x{=}0] = ATT + \text{Selection Bias}$$

(f) Suppose instead that treatment were **randomly assigned** (each individual gets $x = 1$ with probability 0.5, regardless of type). What would $E[\bar{y}_1 - \bar{y}_0]$ estimate?

Block 4: The Selection Bias Decomposition and Random Assignment (13 min)

Question 5 (Key derivation)

We now prove the result we used numerically in Block 3. Let $\bar{y}_1$ be the average outcome among treated units and $\bar{y}_0$ among untreated.

(a) Show that the **population** version of the comparison $\bar{y}_1 - \bar{y}_0$ decomposes as:

$$E[y \mid x{=}1] - E[y \mid x{=}0] = \underbrace{E\big[y(1) - y(0) \mid x{=}1\big]}_{ATT} + \underbrace{E\big[y(0) \mid x{=}1\big] - E\big[y(0) \mid x{=}0\big]}_{\text{Selection Bias}}$$

Hint: use the switching equation to write $E[y \mid x{=}1] = E[y(1) \mid x{=}1]$ and $E[y \mid x{=}0] = E[y(0) \mid x{=}0]$. Then add and subtract $E[y(0) \mid x{=}1]$.

(b) Explain in words what $E[y(0) \mid x{=}1] - E[y(0) \mid x{=}0]$ means. Use the education example (Ex. 1.4) and the job training context to give one example of **positive** and one of **negative** selection bias.

(c) Now suppose treatment is **randomly assigned**. Before doing the math, let us understand *why* randomization works.

Why does random assignment "work"? — The mathematical mechanism

Step 0: A key property of independence. Recall that if $X \perp Y$, then

$$E[Y \mid X = x] = E[Y] \quad \text{for all } x.$$

Conditioning on an independent variable **does nothing**: the conditional expectation equals the unconditional one. *This single fact is the entire engine.*

Step 1: Why does randomization create independence? Before the experiment, each unit i already has *fixed* potential outcomes $(y_i(0), y_i(1))$, determined by that person's characteristics (ability, motivation, health, . . .). These exist regardless of what we do.

When we randomly assign treatment, x_i is determined by a coin flip that **knows nothing** about unit i. Formally:

$$P(x_i = 1 \mid y_i(0),\, y_i(1)) = P(x_i = 1) = p \quad \text{for all } i$$

A person with $y_i(0) = 100$ has the *same* probability p of being treated as a person with $y_i(0) = 2$. The coin does not look at the potential outcomes. Therefore, **by construction**:

$$\boxed{x \perp (y(0),\ y(1))}$$

Step 2: Apply the independence property. Since $x \perp y(0)$:

$$E[y(0) \mid x{=}1] = E[y(0)] \quad \text{and} \quad E[y(0) \mid x{=}0] = E[y(0)]$$

The treated group and the control group have **the same average baseline** — not because they are identical person-by-person, but because the *sorting mechanism* (the coin) is unrelated to individual characteristics. Likewise, $x \perp y(1)$ gives $E[y(1) \mid x{=}1] = E[y(1)]$.

Step 3: Why does this fail without randomization? Without randomization, people choose (or are selected for) treatment based on their characteristics — the same characteristics that determine $(y_i(0), y_i(1))$. So $P(x_i = 1 \mid y_i(0), y_i(1)) \neq P(x_i = 1)$:

the probability of treatment *depends on* potential outcomes. For example, if high-ability people are more likely to go to college, then $E[y(0) \mid x{=}1] > E[y(0) \mid x{=}0]$. Conditioning on $x = 1$ selects a non-representative subgroup, and the independence property fails.

Now use the property $x \perp (y(0), y(1))$ to show three things:

(i) Selection bias $= 0$.

(ii) $ATT = ATE$.

(iii) $\bar{y}_1 - \bar{y}_0$ is unbiased for the ATE.

(d) (Problem 15 from the book) The **sample average treatment effect** estimator is $\hat{\tau}_{ate} = n^{-1}\sum_{i=1}^{n}[y_i(1) - y_i(0)]$. Show that *if* we could observe both potential outcomes for everyone, $\hat{\tau}_{ate}$ would be unbiased for $\tau_{ate} = E[y(1) - y(0)]$.

(e) Explain briefly why $\bar{y}_1$ and $\bar{y}(1) \equiv n^{-1}\sum_{i=1}^{n} y_i(1)$ are *not* the same thing.

Block 5: Application — Job Training Program (6 min)

Question 6 (From Example 2.14 in the book)

Example 2.14 — Evaluating a Job Training Program (JTRAIN2)

The data in JTRAIN2 are from the National Supported Work demonstration, where men were **randomly assigned** to receive job training (9–18 months) or serve as controls. The outcome is `re78`: real earnings in 1978 (thousands of 1982 dollars). The treatment variable is `train` ($= 1$ if trained).

Of 445 men, 185 received training and 260 did not. The simple regression $\widehat{re78} = \hat{\beta}_0 + \hat{\beta}_1 \cdot train$ gives: $\hat{\beta}_0 = 4.555$ (control group average, in thousands), $\hat{\beta}_1 = 1.794$ (treated earned \$1,794 more on average). The t-statistic is 1.79, and $R^2 \approx 0.018$.

The small R^2 means training explains only 1.8% of earnings variation. But R^2 measures **explanatory power**, not causal significance — under random assignment, $\hat{\beta}_1$ is still unbiased for the ATE.

(a) Express $\bar{y}_1$, $\bar{y}_0$, and $\hat{\beta}_1$ in terms of each other. Compute $\bar{y}_1$.

(b) Can we interpret $\hat{\beta}_1 = 1.794$ causally? Why? Express the effect as a percentage of the control mean.

(c) The R^2 is 0.018. A student says: "The training had no effect because R^2 is tiny." Is this correct?

(d) If workers had **volunteered** instead of being randomly assigned, would you still trust the estimate? Why?